Determinants of Global Life Expectancy: A Cross-Country Econometric Analysis

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Research Objective & Data Specification

Objective: To empirically identify the primary macroeconomic and infrastructural drivers of global life expectancy using cross-sectional modeling.

Data Source: World Bank Indicators (2022 Snapshot)

- **Dependent Variable (Y):** Life Expectancy at Birth (Years)
- **Core Regressors (X_k):**
  - Basic Sanitation Access (% of population)
  - GDP per Capita (Constant USD)
  - Urban Population (% of total)

- **Sample Size Strategy ($n = 171$):** Variables with severe reporting deficits (e.g., Literacy Rate, missing 172 observations) were excluded from the primary model. This prevents a massive loss of degrees of freedom, satisfying the large-sample requirement for asymptotic consistency ($\text{plim}_{n\to\infty}\hat{\beta} = \beta$).

Theoretical Framework:

The Income-Health Paradox: It is a biological fact that life expectancy cannot grow infinitely with wealth. To capture this asymptotic ceiling, we specify Income as a reciprocal (hyperbolic) function.

The Core Model:

$$\text{Life_Exp}_i = \beta_0 + \beta_1(\text{Sanitation}_i) + \beta_2\left(\frac{1}{\text{GDP_PC}_i}\right) + \beta_3(\text{Urban_Pop}_i) + \epsilon_i$$

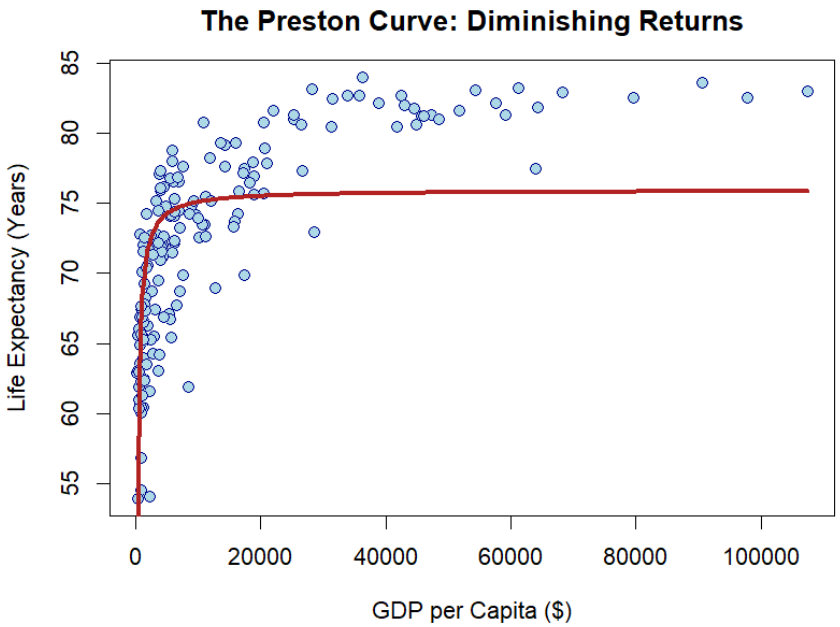
Mathematical Interpretation:

- As GDP_PC $\to \infty$, the penalty term $\left(\frac{1}{\text{GDP_PC}}\right) \to 0$.
- The theoretical maximum life expectancy ceiling is therefore constrained strictly by β_0, β_1, and β_3.
- This inherently models the diminishing marginal returns to national wealth.

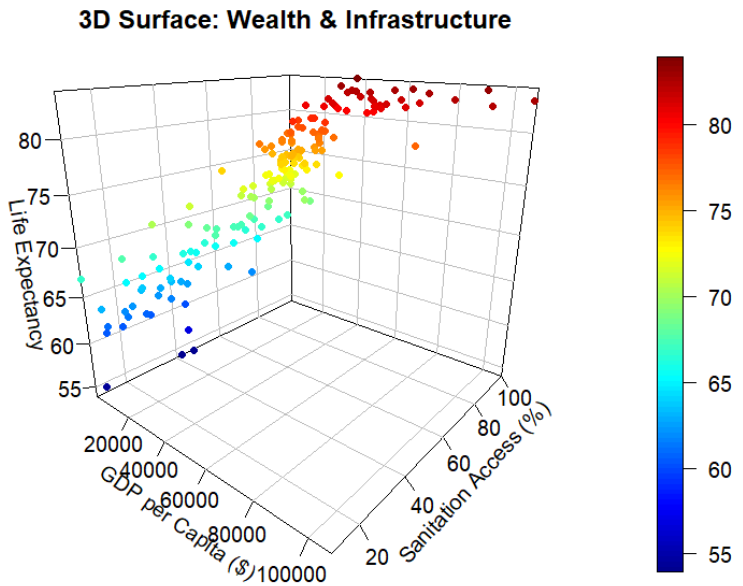
Empirical Justification: 2-D and 3-D visualizations

Here, we visually map the severe structural non-linearities inherent in our cross-country demographic data.

2D - Wealth



3D - Wealth and Sanitation



Structural Inference:

The 2D scatter (left) demonstrates extreme diminishing marginal returns to wealth, empirically justifying our choice of hyperbolic function. The 3D surface (right) incorporates Sanitation, revealing a multi-dimensional biological asymptote where infrastructural capacity dictates the ultimate longevity ceiling.

Baseline Results: The Level Specification

Table: Dependent Variable: Life Expectancy at Birth (Years)

Variable	**Estimate**	**Std. Error**	**p-value**
Intercept ($\hat{\beta}_0$)	57.290***	(2.050)	< 0.001
Sanitation (%) ($\hat{\beta}_1$)	0.175***	(0.021)	< 0.001
Income $I(1/\text{GDP_PC})$ ($\hat{\beta}_2$)	-3040.000**	(925.800)	0.00125
Urban Population (%) ($\hat{\beta}_3$)	0.042*	(0.020)	0.03915
Observations (n)	171		
R^2 / Adjusted R^2	0.669 / 0.663		
F-Statistic	112.600 ($p < 0.001$)		

Note: Significance levels: *p < 0.05; **p < 0.01; ***p < 0.001. Standard OLS errors applied.

Initial Inference: Standard OLS suggests all selected infrastructural and economic variables are statistically significant influencers of life expectancy, capturing roughly 66.9% of the cross-country variations.

The Log Specification & Endogeneity Limitations

The Log-Log & Log-Lin Model:

$$\ln(\text{Life_Exp}_i) = \alpha_0 + \alpha_1 \ln(\text{GDP_PC}_i) + \alpha_2(\text{CO2_PC}_i) + \alpha_3(\text{Health_Exp}_i) + \mu_i$$

Table: Dependent Variable: ln(Life Expectancy)

Variable	**Estimate**	**Std. Error**	**p-value**
Intercept ($\hat{\alpha}_0$)	3.662***	(0.053)	< 0.001
ln(GDP_PC) ($\hat{\alpha}_1$)	0.073***	(0.006)	< 0.001
CO_2 per Capita ($\hat{\alpha}_2$)	-0.001	(0.001)	0.274
Health Exp. (% GDP) ($\hat{\alpha}_3$)	-0.003	(0.002)	0.139
Observations (n)	171		
R^2 / Adjusted R^2	0.473 / 0.463		
F-Statistic	49.91 ($p < 0.001$)		

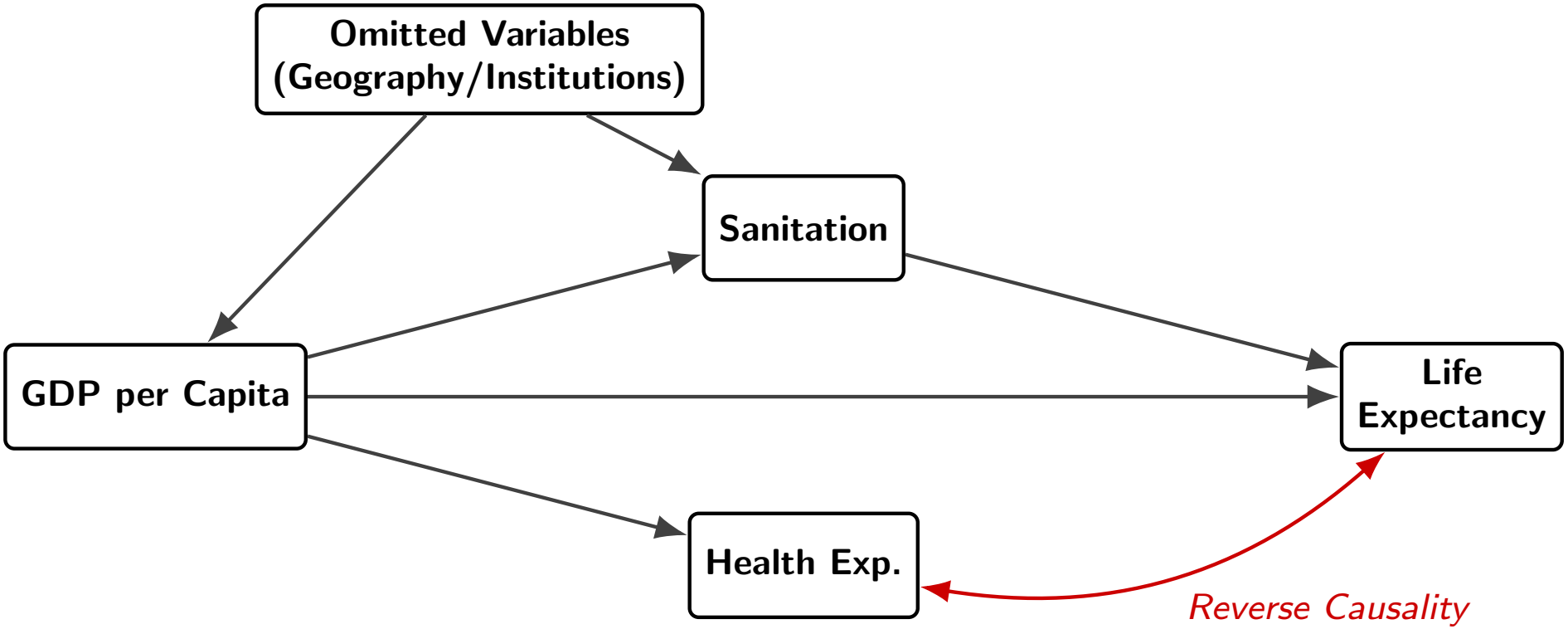
Note: Significance levels: *p < 0.05; **p < 0.01; ***p < 0.001. Standard OLS errors applied.

Econometric Critique: Reverse Causality

The negative coefficient for Health Expenditure highlights a structural flaw. Countries experiencing demographic aging or acute health crises typically allocate a higher share of GDP to healthcare. Because the dependent variable (Y) drives the independent variable (X), $\text{Cov}(X, \epsilon) \neq 0$. This endogeneity renders the estimator inconsistent.

Structural Causal Model & Endogeneity Pathways

The following is the exact graph showing the potential causal loops and endogeneity pathways as we just saw in the previous slide. Please note that the following diagram is NOT exhaustive!



Diagnostic Testing: Multicollinearity

The Threat of Multicollinearity: Economic intuition suggests that affluent countries inherently possess better sanitation and higher urbanization. If these regressors are highly collinear, they artificially inflate standard errors, hiding true statistical significance.

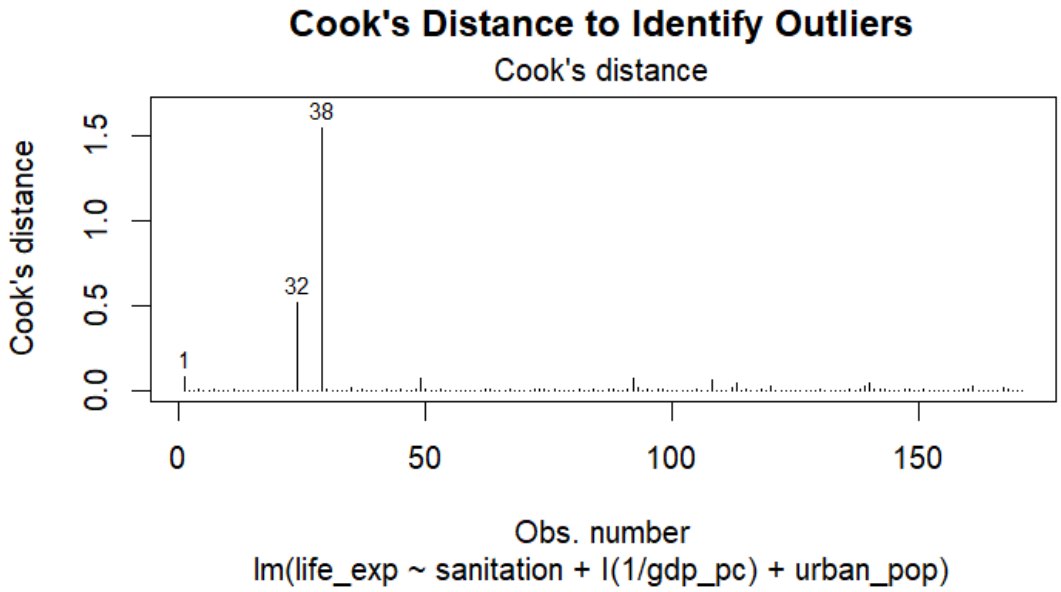
Table: Variance Inflation Factor (VIF) Results

Variable	**VIF Score**
Basic Sanitation (%)	2.289
Income $I(1/\text{GDP_PC})$	2.229
Urban Population (%)	1.517

Conclusion: All VIF scores fall well below the standard danger threshold of 5.0. This confirms that our independent variables are capturing distinct, non-overlapping dimensions of national infrastructure without mathematically conflating each other.

Diagnostic Testing: Outlier Detection

Cook's Distance Analysis: Because cross-country macroeconomic data is highly susceptible to extreme national anomalies, we measure the influence of individual observations. High Cook's Distance values mathematically distort the OLS line of best fit.



Visual Inference:
The plot identifies observations 32 and 38 as extreme outliers. These specific nations exert a huge pull on the regression curve, requiring a targeted robustness check to ensure they are not driving the baseline results.

The Anatomy of Outliers: China and Cuba

Upon isolating the influential observations, we identified two countries fundamentally skewing the mathematical assumptions of our baseline model:

1. **The Cuban Health Paradox**
   - *Data:* Low GDP per capita ($7,552) but first-world life expectancy (77.6 years).
   - *Impact:* Cuba's highly specialized, preventative-care-focused medical system breaks standard macroeconomic income-to-health models, thereby significantly influencing the regression line.
2. **The Scale of China**
   - *Data:* Middle-income GDP ($11,830) paired with an unusually high life expectancy (78.2 years) and near-universal sanitation (95.8%).
   - *Impact:* China's unprecedented scale and rapid demographic transition act as a heavy mathematical anchor, dragging predicted values away from the traditional global trajectory.

Robustness Check: Model Without Outliers

Table: Dependent Variable: Life Expectancy (Excluding China & Cuba)

Variable	**Estimate**	**Std. Error**	**p-value**
Intercept ($\hat{\beta}_0$)	57.300***	(2.061)	< 0.001
Sanitation (%) ($\hat{\beta}_1$)	0.175***	(0.021)	< 0.001
Income I(1/GDP_PC) ($\hat{\beta}_2$)	-3036.000**	(930.900)	0.00135
Urban Population (%) ($\hat{\beta}_3$)	0.042*	(0.020)	0.04082
Observations (n)	169		
R^2 / Adjusted R^2	0.668 / 0.662		
F-Statistic	110.400 ($p < 0.001$)		

Note: Significance levels: *p < 0.05; **p < 0.01; ***p < 0.001. Standard OLS errors applied.

Inference:
Dropping the extreme anomalies of China and Cuba does not fundamentally alter our baseline estimates; both Income and Urbanization maintain strict statistical significance. This confirms our naive model is robust to outliers. However, the critical diagnostic question remains: did these outliers cause the non-constant variance in our error term?

Diagnostic Testing: Heteroskedasticity

The Homoskedasticity Assumption: Standard OLS mathematically requires the variance of the error term to be constant across all observations: $\text{Var}(\epsilon_i|X_i) = \sigma^2$. If this fails, $\hat{\beta}$ remains unbiased, but the standard errors become invalid.

Breusch-Pagan Test (Baseline Model, $n = 171$):

- **Null Hypothesis (H_0):** Homoskedasticity is present.
- **Test Statistic (BP):** 20.017
- **p-value:** 0.0001684

Initial Inference: Because $p \ll 0.05$, we strictly reject the null hypothesis. The baseline model suffers from severe heteroskedasticity. However, the critical question is whether this non-constant variance is being artificially generated by our extreme outliers (China and Cuba).

The Persistent Variance Matrix

To test if the outliers caused the heteroskedasticity, we re-ran the Breusch-Pagan test on our isolated $n = 169$ model (excluding China and Cuba).

Breusch-Pagan Test (Robust Model, $n = 169$):

- **Test Statistic (BP):** 19.701
- **p-value:** 0.0001958

Structural Inference:
Heteroskedasticity is still present ($p \ll 0.05$). This demonstrates that the variance violation is a **structural macroeconomic reality** rather than an outlier anomaly. Life expectancies in developed countries are closely clustered around the biological maximum (low variance), whereas those in developing countries fluctuate greatly because of famines, wars, and different baseline infrastructures (high variance). Our standard errors now need to be adjusted mathematically.

The Econometric Fix: Robust Standard Errors

The Analytical Solution: Because structural heteroskedasticity persists even after outlier removal, the standard OLS variance estimator $\text{Var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$ is mathematically invalid and artificially small.

To compute honest p-values, we apply the Robust Standard Error (HC1) Sandwich Estimator. This recalculates the variance-covariance matrix without altering our unbiased $\hat{\beta}$ coefficients:

$$\text{Var}_{HC1}(\hat{\beta}) = (X'X)^{-1}(X'\Omega X)(X'X)^{-1}$$

- $(X'X)^{-1}$ **relies strictly on the observed data.**
- $(X'\Omega X)$ **accounts for the non-constant variance (heteroskedasticity) of the residuals across different nations.**

Expectation: By applying this fix to our $n = 169$ model, standard errors will structurally inflate, providing a much stricter test of statistical significance for our regressors.

Final Robust Results: The Ultimate Specification

Table: Dependent Variable: Life Expectancy (HC1 Standard Errors, $n = 169$)

Variable	**Estimate**	**Robust Std. Error**	**p-value**
Intercept ($\hat{\beta}_0$)	57.300***	(2.577)	< 0.001
Sanitation (%) ($\hat{\beta}_1$)	0.175***	(0.022)	< 0.001
Income I(1/GDP_PC) ($\hat{\beta}_2$)	-3036.400	(1949.600)	0.1213
Urban Population (%) ($\hat{\beta}_3$)	0.042.	(0.022)	0.0657

Note: Significance levels: . p < 0.1; *p < 0.05; **p < 0.01; ***p < 0.001.

The Econometric Plot Twist:
The standard errors for income and urbanization greatly increase when both structural heteroskedasticity and extreme outliers (China/Cuba) are mathematically taken into consideration. The 5% level of statistical significance of both variables is essentially lost. This thorough diagnostic procedure demonstrates that basic sanitation infrastructure ($p = 1.08 \times 10^{-13}$) is the most significant, mathematically reliable predictor of life expectancy worldwide, greatly outperforming pure economic wealth in this model.

Diagnostic Testing: Misspecification

Ramsey RESET Test: Even with robust standard errors, we must prove our functional form captures the true nature of the data. The Ramsey Regression Equation Specification Error Test (RESET) checks for omitted non-linear variables by adding polynomials of the fitted values ($\hat{Y}^2$, $\hat{Y}^3$) back into the model.

RESET Test Results (Robust Model):

- **Null Hypothesis (H_0):** No functional form misspecification.
- **Test Statistic:** 21.526
- **Degrees of Freedom:** df1 = 2, df2 = 163
- **p-value:** 5.059×10^{-9}

Structural Inference:
Because $p \ll 0.05$, we strongly reject the null hypothesis. The test mathematically indicates missing non-linear relationships. Economically, this implies that variables like sanitation likely exhibit diminishing marginal returns—the jump from 10% to 20% access saves more lives than the jump from 80% to 90%. Future ideal models require quadratic terms (Sanitation^2) to map these biological ceilings.

The Asymptotic Penalty: Small Sample Theory

The Missing Variable Dilemma: Literacy is a theoretically vital determinant of public health. However, 126 nations failed to report this metric in the 2022 World Bank snapshot. To analyze its impact, we must isolate a sub-model, crashing our sample size down to just $n = 45$.

The Variance Inflation Mechanic: In a multivariate OLS framework, the variance of an estimator is defined as:

$$\mathrm{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\sum(X_{ij} - \bar{X}_j)^2} \cdot \mathrm{VIF}_j$$

Theoretical Consequence:
As we massively cut down our sample size, the Total Sum of Squares in the denominator collapses greatly. Dividing by a drastically smaller number artificially inflates the overall variance ($\mathrm{Var}(\hat{\beta}_j)$). This mathematical penalty perfectly demonstrates why preserving degrees of freedom was critical in our primary $n = 171$ specification.

The Literacy Sub-Model: Empirical Collapse

Table: Dependent Variable: Life Expectancy ($n = 45$)

Variable	**Estimate**	**Std. Error**	**p-value**
Intercept ($\hat{\beta}_0$)	32.088***	(3.588)	< 0.001
Literacy Rate (%) ($\hat{\beta}_1$)	0.083.	(0.047)	0.0845
ln(GDP_PC) ($\hat{\beta}_2$)	3.905***	(0.701)	< 0.001
Observations (n)	45		
R^2 / Adjusted R^2	0.761 / 0.750		
F-Statistic	66.91 ($p < 0.001$)		

Note: Significance levels: . p < 0.1; *p < 0.05; **p < 0.01; ***p < 0.001.

Empirical Inference:
As mathematically predicted, shrinking the sample size mechanically inflated the standard error for Literacy (0.047). This crushed the t-statistic, rendering Literacy formally insignificant at the 5% level ($p = 0.0845$) despite its known importance. This confirms the severe asymptotic penalty of missing data.

Advanced Limitations: Spatial Autocorrelation

The Cross-Sectional Constraint: Because this is a 2022 cross-sectional snapshot, traditional time-series autocorrelation tests (like Durbin-Watson) are structurally invalid due to arbitrary alphabetical ordering. However, our model remains highly vulnerable to **Spatial Autocorrelation**.

Geographic Error Clustering:

- Health crises, pollution clouds, and economic shocks aren't naturally confined within the geographical borders of the countries.
- For instance, the unobserved error term (ϵ) for France is highly likely to be correlated with the error term for Germany: $\text{Cov}(\epsilon_i, \epsilon_j) \neq 0$ for neighboring nations i and j.

Future Modeling Fixes:
To formally test for geographic dependence, future extensions of the model must be upgraded to a Spatial Autoregressive (SAR) or Spatial Error Model (SEM) to protect the validity of the standard error.

Conclusion & Policy Implications

1. The Diagnostic Reality Standard OLS created a naive illusion of perfect macroeconomic drivers. Rigorous testing revealed severe structural heteroskedasticity and extreme outlier distortion (China and Cuba), fundamentally invalidating our baseline variance assumptions.

2. The Robust Verdict Applying HC1 standard errors mathematically corrected the variance matrix. This strict correction stole from global income, its 5% statistical significance, successfully isolating **basic sanitation infrastructure** as the true, robust driver of cross-country life expectancy.

3. Macroeconomic Policy Imperative

- *The Infrastructure Focus:* General income growth does not automatically translate to longer lives. Direct state investment in broad public goods provides a guaranteed, measurable return on public health.
- *Diminishing Marginal Returns:* As hinted by our non-linear diagnostics, infrastructure investments yield the highest dividends at the absolute bottom of the development curve.

Thank You.
Questions?